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    if admission_register_list  
    for program
```

**DESIGN AND BUILD WEB APPLICATIONS, DESKTOP
GRAPHICAL USER INTERFACES OR MOBILE APPS, NQF LEVEL**

5, CREDITS 8

PRACTICAL MODULE

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Module #	251201-005-00-PM-09:
NQF Level	level 5
Notional hours	80
Credit(s)	Cr 8
Occupational Code	251201005
Qualification Title	Occupational Certificate: Software Developer

Introduction

As part of your training you are required to keep a logbook of all practical on the job training and exposure you receive during the learning process.

You will now be assigned to a mentor who will oversee your off-site training, usually referred to as on-the-job training.

The mentor will assist and advise you on the practical aspects of the job, how to fit into the company, what is expected of you as an employee and as a future supervisor.

Responsibilities of the learner include:

- One hundred percent commitment to the learning process. Learners are encouraged to study any additional source of information relevant to this learning process.
- Doing all assignments contained in this workbook as well any tasks and assignments received from your mentor or supervisor to whom you have been assigned.
- Although the mentor is responsible to sign off all sections completed, it is the learner's responsibility to ensure that all paperwork is completed and handed in for filing on his/her record of learning. It should be clearly stated to learners that a 100% complete record of learning, as prescribed by this logbook, is **their sole responsibility**. Any document missing from the record may result in your not being declared competent.
- Discuss any problems that you may have with your mentor.

Acknowledgment of Receipt

I Lindokuhle Pearlson Molo (Learner) acknowledge receipt of my on the job Training Logbook on this the 11th day of May 2026. The process of on-the-job training has been explained to me.

A handwritten signature in black ink, appearing to be 'Lindokuhle Pearlson Molo', written in a cursive style.

Signature of Learner

Name of Mentor: Nicholas Govender

Signature of Mentor

Declaration of Authenticity

I (*learner name and surname*) Lindokuhle Pearlson Moloj, ID Nr 0003055567089 hereby declare that the work contained herein was completed by me on my own.

Where assistance or advice was received or where I used resource material from a workbook, policy wording, internet, or any other printed sources, this has been acknowledged and referenced. I further declare that I understand that plagiarism is a punishable offence as it constitutes the theft of another's intellectual property rights.



11 May 2026

Learner signature

Date

Declaration by mentor

I (*mentor name and surname*) _____, ID Nr _____ hereby declare that the learner is being mentored by myself and that the functions listed and the working hours is a true reflection of the learner's situation. According to my knowledge I declare that this is his/her own work.

Instructions to the Mentor

Congratulations on your appointment as a mentor to the learner. This is a very responsible assignment because you have been tasked with the responsibility of rounding off the learner's practical exposure.

You must ensure that you are familiar with all aspects of the work covered in this logbook because you must keep a daily account of the learner's performance.

You are also required to report to the learnership coordinator skills development facilitator, or as agreed between yourself, the coordinator and the learner regarding the learner's progress. Your responsibilities as mentor are as follows:

- Study the logbook and acquaint yourself with its content and format
- Remember this logbook is the learner's full record of learning and workplace exposure. All activities which the learner participates in must be recorded, and all documents produced in relation to this learnership must form part of the record of learning
- Get all the learners together and explain its purpose to them and also what is required of them
- Remember the mentor is the creator of learning and exposure opportunities. You should therefore not confine the learner's exposure to this logbook alone
- Continuously guide them in doing the assignments and arranging the planned exposure with the relevant departments
- File all duplicate records of learning on a file for each learner
- Send all original records to the training provider at the end of end of each month. The Seta also requires that copies be held at the companies
- This logbook need not be followed chronologically, but please note that the learner is required to work through the entire logbook by the end of the learning period

Responsibilities of the Employer

- Creating an atmosphere conducive to learning

- Giving learners ample access to the working environment. Remember that learners should be productive employees to get practical exposure to all aspects of the transport operation as required by the learnership
- Ensure that learners, mentors and assessors attend all training required and arrange and pay travelling and accommodation costs
- Ensure availability of sufficient mentors and workplace assessors

Training Provider Responsibility

- Provide all practical learning material in electronic and/or hard copy, depending on the circumstances
- Provide training for mentors and workplace assessors if required and provide the learning material
- Visit employers to monitor progress and provide guidance and feedback.
- Provide an online and telephonic support system to all mentors and learners

Provider Accreditation Requirements for the Module

Provider Accreditation Requirements:

Physical Requirements:**Demonstrate access to:**

- Engineering tools
- Work space and bench per learner
- Fabrication material
- Standard operating procedures
- Original Equipment Manufacturer manuals and specifications

Human Resource Requirements:

- Qualified fitter and turner with 3 years of experience in the trade for every 2 learners
- Recognised expertise in mentorship
- Recognised expertise in assessment Legal Requirements:
- Compliance to the relevant occupational health, safety and environmental protection legislation.

Additional Assignments to be Assessed Externally

- None specified

PRACTICAL SKILL MODULE SPECIFICATIONS

List of Practical Skill Module Specifications

- 251201005-PM-01, Use Software to Communicate and Visualise Information, NQF Level 4, Credits 3
- 251201005-PM-02, Use and Manage Spreadsheets and Workbooks, NQF Level 4, Credits 3
- 251201005-PM-03, Use Desktop Applications to Analyse, Visualise and report on Data, NQF Level 5, Credits 3
- 251201005-PM-04, Use a Visual Analytics Platform and Visualisation Tools to Analyse, Visualise and report on Data, NQF Level 5, Credits 3
- 251201005-PM-05, Query and Massage Data, NQF Level 5, Credits 3
- 251201005-PM-06, Apply Logical Thinking and Maths, NQF Level 4, Credits 3
- 251201005-PM-07, Apply Code to use a Software Toolkit/Platform in the Field of Study or Employment, NQF Level 5, Credits 3
- 251201005-PM-08, Develop Software using HTML5, Opensource Frameworks and Libraries, NQF Level 5, Credits 16
- 251201005-PM-09, Design and Build Web Applications, Desktop Graphical User Interfaces or Mobile Apps, NQF Level 5, Credits 8
- 251201005-PM-10, Use a Cloud Automation Platform to Create Solutions, NQF Level 4, Credits 8
- 251201005-PM-11, Develop Software using Python, NQF Level 5, Credits 12
- 251201005-PM-12, Apply the Development Cycle when Developing Software, NQF Level 5, Credits 16
- 251201005-PM-13, Participate in a Design Thinking for Innovation Workshop, NQF Level 4, Credits 4
- 251201005-PM-14, Function Ethically and Effectively in a Team, NQF Level 4, Credits 4

PURPOSE OF THE PRACTICAL SKILL MODULE 9

Purpose of the Practical Skill Modules

The focus of the learning in this module is on providing the learner with an opportunity to design and create web applications, mobile apps and graphical user interfaces (GUIs), using the **Model-View-Controller (MVC)** framework as an architectural pattern and to separate an application into three main logical components Model, View, and Controller

The learner will be required to:

- PM-09-PS01 Plan a project to build a solution which effectively solve the customer's business problems (project design phase)
- PM-09-PS02 Configure middleware's and services
- PM-09-PS03 Develop controllers for processing web requests
- PM-09-PS04 Develop views to define the user interface for web applications
- PM-09-PS05 Create code to develop MVC models to interact and model various types of data or objects
- PM-09-PS06 Connect an application to a database to access and store data using an object-database mapper to build a database-driven website in MVC
- PM-09-PS07 Build web applications apply a consistent look and feel to the application
- PM-09-PS08 Develop the client-side of a web application using applicable tools
- PM-09-PS09 Test and troubleshoot for bugs that results in exceptions or unexpected behaviour
- PM-09-PS10 Manage security aspects of a web application
- PM-09-PS11 Increase the speed of data access and communication to improve performance
- PM-09-PS12 Implement web APIs to enable and promote application interaction with external systems
- PM-09-PS13 Host and deploy an web application ensuring it is accessible by clients on a wide variety of machines

SECTION A- PM-09-PS01

PM-09-PS01: Plan a project to build a solution which effectively solve the customer's business problems (project design phase)

Scope of Practical Skill

Given a simulated environment where the conditions are typical of those in a working environment including programming standards and guidelines, programming software, required hardware and its components, industry standard software development tools and an integrated development environment (IDE), the learner must be able to:

- PA 0101 Interrogate the business problem to reach a complete understanding of the purposes of a project
- PA 0102 Identify the set of business needs
- PA 0103 Plan the Model-View-Controller (MVC) web application to meet those needs:
 - Plan model classes
 - Plan controllers
 - Plan views
 - Architecting and MVC Web Application
- PA 0104 Plan the overall architecture of an MVC web application and consider aspects such as state management.
- PA 0105 Plan the models, controllers, and views that are required to implement a given set of functional requirements
- PA 0106 Create a project plan which demonstrates an accurate understanding of the requirements
- PA 0107 Include information that communicates the functionality of the web application, its user interface, structure, and data storage to the developers
- PA 0108 Write a detailed and accurate project plan to ensure that the powerful features of MVC are used effectively to solve the customer's business problems

Applied Knowledge

- AK 0101 MVC as an architectural pattern and model
- AK 0102 Features and application of MVC
- AK 0103 The variety of technologies available
- AK 0104 Different programming methodologies
- AK 0105 Describe the role of MVC in the web technologies stack, and how to use MVC to build web applications
- AK 0106 Distinguish between MVC models, MVC controllers, and MVC views

Internal Assessment Criteria

- IA C0101 A detailed and accurate project plan which ensures that the powerful features of MVC are used effectively to solve the customer's business problems is written

Instructions

You are required to perform the task below and your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

NAME: <u>Lindokuhle</u> SURNAME: <u>Moloi</u> COMPANY: <u>Progression</u> ID: <u>0003055567089</u>		DATE: <u>11 May 2026</u> TIME: <u>09:00</u>	
EVALUATION CRITERIONS	COMPETEN T	NOT YET COMPETEN T	COMMENTS OR ACTION REQUIRED
Ability of the learner to:			
PA 0101: Task: Conduct an in-depth meeting with stakeholders to understand the core business problem, gathering relevant documentation to clarify the purpose and expected outcomes of the project.			
PA 0102: Task: Identify and list the key business needs for the			

web application by interviewing stakeholders and analyzing business process workflows.			
• PA 0103: Task: Create a plan for the MVC web application by outlining model classes (representing the business data), controllers (handling business logic and user input), and views (the user interface). Include an initial wireframe for the views and a diagram of the controller flow.			
• PA 0104: Task: Design the overall architecture of the MVC web application, detailing how the application will manage state across multiple requests (e.g., using session management or tokens for authentication).			
• PA 0105: Task: Draft a set of functional requirements for the web application, then outline the specific models, controllers, and views that will be required to meet each of those requirements.			

• PA 0106: Task: Develop a project timeline that includes all stages of the software development lifecycle (SDLC), ensuring that the MVC structure aligns with the customer's business needs and development milestones.			
• PA 0107: Task: Create a comprehensive document that explains the web application's user interface, data storage schema, and structure. Include diagrams of the MVC workflow to communicate functionality to developers.			
• PA 0108: Task: Write a detailed project plan that outlines how the MVC framework will be utilized to meet the customer's business requirements, emphasizing the powerful features such as data-binding, validation, and routing to ensure scalability and maintainability.			

SECTION B- PM-09-PS02

PM-09-PS02: Configure middlewares and services

Scope of Practical Skill

Given a simulated environment where the conditions are typical of those in a working environment including programming standards and guidelines, programming software, required hardware and its components, industry standard software development tools and an integrated development environment (IDE), the learner must be able to:

- PA 0201 Use existing middleware
- PA 0202 Create own middleware and use it to define custom behaviour
- PA 0203 Configure middlewares
- PA 0204 Configure services
- PA 0205 Work with static files
- PA 0206 Create custom middleware
- PA 0207 Use dependency injection
- PA 0208 Inject a service to a controller

Applied Knowledge

- AK 0201 Understand the basic principles behind Dependency Injection, and how it is used
- AK 0202 Know how to create a custom service, configure its scope, and inject it to both middleware and controllers

Internal Assessment Criteria

- IA C0201 Handle requests and responses via existing, and custom middleware
- IA C0202 Configure services for use in middleware and throughout other parts of the application, such as controllers
- IA C0203 Configure middleware to use as part of the request and response pipeline that allows us to handle them according to any relevant parameter
- IA C0204 Use dependency injection

Instructions

You are required to perform the task below and your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

NAME: <u>Lindokuhle</u> SURNAME: <u>Moloi</u> COMPANY: <u>Progression</u> ID: <u>0003055567089</u>		DATE: <u>11 May 2026</u> TIME: <u>09:00</u>	
EVALUATION CRITERIONS	COMPETEN T	NOT YET COMPETEN T	COMMENTS OR ACTION REQUIRED
Ability of the learner to:			
PA 0201: Task: Integrate existing middleware (e.g., authentication or logging middleware) into a pre-built web application, ensuring that it functions according to the application's requirements.			
PA 0202: Task: Develop a custom middleware that logs request and response times for each HTTP request,			

then integrate it into an existing web application to monitor performance.			
• PA 0203: Task: Configure a set of middlewares (e.g., error handling, authentication, logging) in the application's request pipeline to ensure they are executed in the correct order for optimal functionality.			
• PA 0204: Task: Configure services (e.g., database services or email services) in the application's dependency injection container, ensuring they are properly registered for use across the application.			
• PA 0205: Task: Serve static files (such as CSS, JavaScript, or images) from a designated folder in the web application by configuring the application to correctly handle static file requests.			
• PA 0206: Task: Create a custom middleware that enforces IP			

filtering, only allowing requests from specific IP addresses, and integrate it into an existing web application.			
• PA 0207: Task: Implement dependency injection by configuring a service (e.g., a logging or data access service) and injecting it into multiple controllers in the web application to reduce tight coupling.			
• PA 0208: Task: Inject a pre-configured service (e.g., a user authentication service) into a controller and use it to authenticate users based on their login credentials.			

SECTION C- PM-09-PS03:

PM-09-PS03 : Develop controllers for processing web requests

Scope of Practical Skill

Given a simulated environment where the conditions are typical of those in a working environment including programming standards and guidelines, programming software, required hardware and its components, industry standard software development tools and an integrated development environment (IDE), the learner must be able to:

- PA 0301 Write controllers and actions
- PA 0302 Write action filters
- PA 0303 Add controllers and actions to an MVC application
- PA 0304 Configure routes by using the routing table
- PA 0305 Configure routes using attributes
- PA 0306 Add an action filter
- PA 0307 Add a controller to a web application that responds to user actions that are specified in the project design
- PA 0308 Add routes to the routing engine and ensure that URLs are user-friendly in an MVC web application
- PA 0309 Write code in action filters that runs before or after a controller action

Applied Knowledge

- AK 0301 Understanding how controllers work
- AK 0302 Classes, methods, actions
- AK 0303 Uniform Resource Locator (URL) routing

Internal Assessment Criteria

- IA C0301 Define URL mapping rules within the applications
- IA C0302 Maximize the reuse of code in controllers, by writing action filters

- IA C0303 Use action filters to run code before or after every action in your web application, on every action in a controller, or on other combinations of controller actions

Instructions

You are required to perform the task below and your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

NAME: <u>Lindokuhle</u> SURNAME: <u>Moloi</u> COMPANY: <u>Progression</u> ID: <u>0003055567089</u>		DATE: <u>11 May 2026</u> TIME: <u>09:00</u>	
EVALUATION CRITERIONS	COMPETEN T	NOT YET COMPETEN T	COMMENTS OR ACTION REQUIRED
Ability of the learner to:			
PA 0301: Task: Write a controller with basic CRUD (Create, Read, Update, Delete) actions for managing user data in the web application. Ensure each action correctly handles HTTP requests and responses.			
PA 0302: Task: Create a custom action filter that logs the			

execution time of each action in the controller. Apply the filter to an existing controller.			
PA 0303: Task: Add a new controller and its corresponding actions to an existing MVC application. Ensure the actions respond to HTTP GET and POST requests and return the appropriate view.			
PA 0304: Task: Configure the routing table to define custom routes for a set of actions in the controller. Ensure that the routes map clean URLs to the correct controller actions.			
PA 0305: Task: Use attribute-based routing to map a controller action to a custom URL, ensuring the route reflects a specific resource hierarchy (e.g., /products/{id}).			
PA 0306: Task: Add a custom action filter to a controller that			

checks for user authentication before allowing access to the controller's actions.			
• PA 0307: Task: Create a controller that responds to user actions, such as form submissions or button clicks, as specified in the project design. Ensure the controller interacts with the model to process user input.			
• PA 0308: Task: Add custom routes to the routing engine to ensure that all URLs in the MVC application are user-friendly and easily readable (e.g., using /blog/posts instead of /blog?id=123).			
• PA 0309: Task: Write code in an action filter that logs activity before and after the execution of a controller action, such as recording the start and end times of the action.			

SECTION D- PM-09-PS04

PM-09-PS04 : Develop views to define the user interface for web applications

Scope of Practical Skill

Given a simulated environment where the conditions are typical of those in a working environment including programming standards and guidelines, programming software, required hardware and its components, industry standard software development tools and an integrated development environment (IDE), the learner must be able to:

- PA 0401 Create Views
- PA 0402 Use HTML helpers and tag helpers
- PA 0403 Reuse code in views
- PA 0404 Add views to an MVC Application
- PA 0405 Add a partial view
- PA 0406 Add a view component
- PA 0407 Create an MVC view to display data to users
- PA 0408 Use HTML helpers and tag helpers in a view
- PA 0409 Reuse markup in multiple locations throughout an application

Applied Knowledge

- AK 0401 How to write the HTML markup and C# code and use the various helper classes that are built into MVC
- AK 0402 How to create partial views and view components

Internal Assessment Criteria

- IA C0401 Develop views to define the user interface for web applications

Instructions

You are required to perform the task below and your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

NAME: <u>Lindokuhle</u> SURNAME: <u>Moloi</u> COMPANY: <u>Progression</u> ID: <u>0003055567089</u>		DATE: <u>11 May 2026</u> TIME: <u>09:00</u>	
EVALUATION CRITERIONS	COMPETEN T	NOT YET COMPETEN T	COMMENTS OR ACTION REQUIRED
Ability of the learner to:			
PA 0401: Task: Create a view in the MVC application that displays a list of products from the model. Ensure the view is well-structured and includes basic HTML for formatting.			
PA 0402: Task: Use HTML helpers (e.g., @Html.TextBoxFor, @Html.ActionLink) and tag helpers (e.g., <input			

asp-for="Model.Property">) to create a form that allows users to submit data in the view.			
• PA 0403: Task: Reuse common code in multiple views by creating a layout page (e.g., for a site's header and footer) and applying it across the application's views.			
• PA 0404: Task: Add a new view to the MVC application to display user profile information. Ensure the view is properly linked to the corresponding action in the controller.			
• PA 0405: Task: Add a partial view to an existing page that displays a list of related products. Ensure the partial view is reusable across multiple pages of the application.			
• PA 0406: Task: Create and add a view component that dynamically displays recent user activity on the			

website. Ensure the view component is properly integrated into the layout.			
• PA 0407: Task: Create an MVC view that displays customer order details. The view should fetch data from the model and present it in a user-friendly table format.			
• PA 0408: Task: Use HTML helpers and tag helpers to build a form in the view that captures user registration details. Ensure that form validation is supported by the helpers.			
• PA 0409: Task: Reuse common markup, such as a navigation bar, by including it in a layout or as a partial view and applying it to multiple pages in the application.			

SECTION E- PM-09-PS05:

PM-09-PS05 : Create code to develop MVC models to interact and model various types of data or objects

Scope of Practical Skill

Given a simulated environment where the conditions are typical of those in a working environment including programming standards and guidelines, programming software, required hardware and its components, industry standard software development tools and an integrated development environment (IDE), the learner must be able to:

- PA 0501 Creating MVC models
- PA 0502 Working with forms
- PA 0503 Validate MVC Application
- PA 0504 Add a model to an MVC application and write code in it to implement the business logic
- PA 0505 Working with forms
- PA 0506 Add validation
- PA 0507 Use display and edit data annotations
- PA 0508 Validate user input with data annotations

Applied Knowledge

- AK 0501 Models, classes, objects and business logic

Internal Assessment Criteria

- IA C0501 Create an MVC model to model the data for your web application
- IA C0502 Create a model class for each type of object that describes the properties of each type of object and can include business logic that matches business processes

Instructions

You are required to perform the task below and your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

NAME: <u>Lindokuhle</u> SURNAME: <u>Moloi</u> COMPANY: <u>Progression</u> ID: <u>0003055567089</u>		DATE: <u>11 May 2026</u> TIME: <u>09:00</u>	
EVALUATION CRITERIONS	COMPETEN T	NOT YET COMPETEN T	COMMENTS OR ACTION REQUIRED
Ability of the learner to:			
PA 0501: Task: Create a model class for managing customer data, including properties such as Name, Email, and PhoneNumber. Ensure the model is properly structured to interact with the database.			
PA 0502: Task: Build a form in the MVC application that allows users to submit feedback. Ensure the form			

captures user input and maps it to the corresponding model properties.			
• PA 0503: Task: Implement validation logic across the MVC application to ensure that all required fields in the forms are filled out and correct before allowing submission (e.g., ensuring the email format is valid).			
• PA 0504: Task: Add a Product model to the MVC application and write business logic in the model class to calculate discount prices based on product categories.			
• PA 0505: Task: Create a form that allows users to register an account, including fields for username, password, and confirmation. Ensure the form correctly maps to the User model.			
• PA 0506: Task: Add server-side validation to the registration			

form, ensuring that the username is unique, the password meets complexity requirements, and all required fields are populated.			
• PA 0507: Task: Use data annotations such as [Display(Name = "Full Name")] to customize how fields in the User model are displayed and [Editable(false)] to prevent specific fields from being edited in forms.			
• PA 0508: Task: Add data annotations like [Required], [EmailAddress], and [StringLength] to the model properties to validate user input, ensuring proper error messages are shown when validation fails.			

SECTION F- PM-09-PS06:

PM-09-PS06 : Connect an application to a database to access and store data using an object-database mapper to build a database-driven website in MVC

Scope of Practical Skill

Given a simulated environment where the conditions are typical of those in a working environment including programming standards and guidelines, programming software, required hardware and its components, industry standard software development tools and an integrated development environment (IDE), the learner must be able to:

- PA 0601 Introduction to object-database mappers
- PA 0602 Work with object-database mappers
- PA 0603 Use an object-database mapper to connect to SQL Server
- PA 0604 Add object-database mapper
- PA 0605 Use object-database mapper to retrieve and store data
- PA 0606 Use object-database mapper to connect to SQL Server
- PA 0607 Separate business logic from data access logic by using a repository. A repository is a class that a controller can call to read data from a data store and to write data to a data store
- PA 0608 Connect an application to a database to access and store data

Applied Knowledge

- AK 0601 Explain functions of an object-database mapper
- AK 0602 data store, repositories,

Internal Assessment Criteria

- IA C0601 Connect an application to a database to access and store data using an object-database mapper
- IA C0602 Render webpages that use data from a data store, to create a web application that changes continually in response to user input, administrative actions, and publishing events
- IA C0603 In Model-View-Controller (MVC) applications, you can create a model that implements data access logic and business logic

Instructions

You are required to perform the task below and your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

NAME: <u>Lindokuhle</u> SURNAME: <u>Moloi</u> COMPANY: <u>Progression</u> ID: <u>0003055567089</u>		DATE: <u>11 May 2026</u> TIME: <u>09:00</u>	
EVALUATION CRITERIONS	COMPETEN T	NOT YET COMPETEN T	COMMENTS OR ACTION REQUIRED
Ability of the learner to:			
PA 0601: Task: Research and write a brief introduction on object-database mappers (ORMs), explaining their role in simplifying data interactions between an application and a database.			
PA 0602: Task: Set up an object-database mapper (e.g., Entity			

Framework) in an MVC application and use it to create a database from an existing model class.			
• PA 0603: Task: Use an object-database mapper like Entity Framework to connect an MVC application to a SQL Server database and retrieve a list of products.			
• PA 0604: Task: Add an object-database mapper to an existing MVC application, ensuring that it is correctly configured to map model classes to database tables.			
• PA 0605: Task: Use the object-database mapper to retrieve a list of customers from the database and display the data in a view. Implement functionality to store new customer records in the database.			
• PA 0606: Task: Use the object-database mapper to establish a connection to SQL Server and retrieve product data for display in an MVC web application.			

• PA 0607: Task: Create a repository class that separates business logic from data access logic. Use the repository pattern to allow a controller to interact with the database for reading and writing customer data.			
• PA 0608: Task: Connect the MVC application to a SQL Server database and create basic functionality for storing and retrieving data using the object-database mapper.			

SECTION G- PM-09-PS07

PM-09-PS07 : Build web applications apply a consistent look and feel to the application

Scope of Practical Skill

Given a simulated environment where the conditions are typical of those in a working environment including programming standards and guidelines, programming software, required hardware and its components, industry standard software development tools and an integrated development environment (IDE), the learner must be able to:

- PA 0701 Apply a consistent layout and link views to it
- PA 0702 Include consistent header and footer sections in all the views
- PA 0703 Use cascading style sheets, styles and layouts that enhance the appearance and usability of the web application
- PA 0704 Create interactive HTML elements
- PA 0705 Use libraries

Applied Knowledge

- AK 0701 Add JavaScript code to your web application.
- AK 0702 Use the jQuery library in your web application.
- AK 0703 Using Layouts
- AK 0704 Using CSS and JavaScript
- AK 0705 Using jQuery

Internal Assessment Criteria

- IA C0701 Build web applications apply a consistent look and feel to the application

Instructions

You are required to perform the task below and your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

NAME: <u>Lindokuhle</u> SURNAME: <u>Moloi</u> COMPANY: <u>Progression</u> ID: <u>0003055567089</u>		DATE: <u>11 May 2026</u> TIME: <u>09:00</u>	
EVALUATION CRITERIONS	COMPETEN T	NOT YET COMPETEN T	COMMENTS OR ACTION REQUIRED
Ability of the learner to:			
PA 0701: Task: Create a consistent layout page that includes common elements like the navigation bar and sidebar. Link multiple views in the MVC application to this layout, ensuring uniformity across pages.			
PA 0702: Task: Add consistent header and footer sections to all views in the application by including them in the			

layout page. Ensure that the header contains the site logo and navigation links, while the footer includes contact information and social media links.			
• PA 0703: Task: Use cascading style sheets (CSS) to define a set of styles (e.g., fonts, colors, spacing) and apply them throughout the application to create a visually appealing and user-friendly interface.			
• PA 0704: Task: Create interactive HTML elements, such as a dropdown menu or an accordion, to enhance user experience. Ensure the elements are responsive and fit seamlessly with the rest of the application.			
• PA 0705: Task: Integrate a front-end library (e.g., Bootstrap or jQuery) into the application to streamline the creation of responsive and interactive web elements. Use the library to add a responsive navigation bar and modal dialogs.			

SECTION H- PM-09-PS08

PM-09-PS08 : Develop the client-side of a web application using applicable tools

Scope of Practical Skill

Given a simulated environment where the conditions are typical of those in a working environment including programming standards and guidelines, programming software, required hardware and its components, industry standard software development tools and an integrated development environment (IDE), the learner must be able to:

- PA 0801 Apply styles
- PA 0802 Apply responsive design tools to customize the web application's display based on the capabilities and specifications of a web browser or a device
- PA 0803 Set up and use task runners to run tasks and compile pre-processor files
- PA 0804 Use a task runner to perform bundling and minification of CSS and files
- PA 0805 Set up a watcher task
- PA 0806 Use Bootstrap framework to style a web application
- PA 0807 Apply CSS flexbox layout
- PA 0808 Write CSS media queries
- PA 0809 Use CSS pre-processors to add features to CSS, such as variables, nested rules, and functions
- PA 0810 Ensure that a web application displays correctly on devices with different screen sizes

Applied Knowledge

- AK 0801 Styles
- AK 0802 Task runners
- AK 0803 Responsive design
- AK 0804 Pre-processors

Internal Assessment Criteria

- IA C0801 Ensure that a web application displays correctly on devices with different screen sizes.
- IA C0802 Use tools to improve the maintainability of complex web applications

Instructions

You are required to perform the task below and your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

NAME: <u>Lindokuhle</u> SURNAME: <u>Moloi</u> COMPANY: <u>Progression</u> ID: <u>0003055567089</u>		DATE: <u>11 May 2026</u> TIME: <u>09:00</u>	
EVALUATION CRITERIONS	COMPETEN T	NOT YET COMPETEN T	COMMENTS OR ACTION REQUIRED
Ability of the learner to:			
PA 0801: Task: Apply custom CSS styles to buttons, forms, and text elements in a web application to enhance its visual appeal and consistency.			
PA 0802: Task: Implement responsive design by using CSS media queries to adjust the layout of a web			

application for different screen sizes (e.g., mobile, tablet, desktop).			
• PA 0803: Task: Set up a task runner like Gulp or Grunt to automate common tasks such as compiling Sass/LESS files and refreshing the browser during development.			
• PA 0804: Task: Use a task runner (e.g., Webpack or Gulp) to bundle and minify CSS and JavaScript files to improve web application performance by reducing file size and load time.			
• PA 0805: Task: Set up a watcher task in a task runner to automatically recompile CSS or JavaScript files when changes are detected, ensuring the latest versions are applied in real time.			
• PA 0806: Task: Use the Bootstrap framework to style and structure the web application. Implement			

Bootstrap's grid system to create a responsive layout and use pre-built components like cards and navigation bars.			
• PA 0807: Task: Apply CSS Flexbox to create a flexible layout that adjusts and distributes space between elements in a web application, ensuring elements are properly aligned and spaced across different devices.			
• PA 0808: Task: Write CSS media queries to adjust font sizes, margins, and layout for different devices, such as scaling down text and repositioning elements on smaller screens.			
• PA 0809: Task: Use a CSS pre-processor like Sass or LESS to add features such as variables for colors and fonts, nested rules, and functions to streamline your styling workflow.			

• PA 0810: Task: Test and ensure that the web application displays correctly on various devices with different screen sizes (mobile phones, tablets, desktops) and fix any layout or styling issues.			
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SECTION I- PM-09-PS09:

PM-09-PS09 : Test and troubleshoot for bugs that results in exceptions or unexpected behaviour

Scope of Practical Skill

Given a simulated environment where the conditions are typical of those in a working environment including programming standards and guidelines, programming software, required hardware and its components, industry standard software development tools and an integrated development environment (IDE), the learner must be able to:

- PA 0901 Test MVC applications throughout the project life cycle to improve quality and ensure that there are no bugs in production software
- PA 0902 Run tests on small components of the web application to ensure that they function as expected
- PA 0903 Test a model
- PA 0904 Test a controller using a fake repository
- PA 0905 Implement a repository in MVC project
- PA 0906 Implement an exception handling strategy
- PA 0907 Handle exceptions and unexpected occurrences while they occur
- PA 0908 Use logs throughout the application:
 - Monitor user activities that might lead to unexpected issues and find solutions to bugs, which you normally would be unsure how to reproduce, by following flows which occurred on the production environment and finding additional errors
- PA 0909 Run unit tests against the Model–View–Controller (MVC) components, such as model classes and controllers, and locate potential bugs
- PA 0910 Build an MVC application that handles exceptions smoothly and robustly
- PA 0911 Run logging providers that benefit your applications and run them by using a common logging API

Applied Knowledge

- AK 0901 Testing tools, methods and techniques
- AK 0902 Troubleshooting tools, methods and techniques
- AK 0903 Debugging tools, methods and techniques

Internal Assessment Criteria

- IA C0901 Identify bugs from any source and eliminate them before you assemble them into a complete web application
- IA C0902 The implications of incorrect assumptions, inaccurate understanding, coding errors, in terms of creating bugs are understood

Instructions

You are required to perform the task below and your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

NAME: <u>Lindokuhle</u> SURNAME: <u>Moloi</u> COMPANY: <u>Progression</u> ID: <u>0003055567089</u>		DATE: <u>11 May 2026</u> TIME: <u>09:00</u>	
EVALUATION CRITERIONS	COMPETEN T	NOT YET COMPETEN T	COMMENTS OR ACTION REQUIRED
Ability of the learner to:			
PA 0901: Task: Test an MVC application throughout its life cycle by implementing unit tests at each stage of development. Run tests after each feature addition to ensure that the application remains bug-free before moving to production.			
PA 0902: Task: Write and run unit tests on small components			

of the web application, such as a form validation method, to ensure they function as expected in isolation.			
• PA 0903: Task: Test a model class by writing unit tests that validate its methods and ensure that the data it handles conforms to business rules (e.g., calculating product prices correctly).			
• PA 0904: Task: Use a fake repository to mock data interactions in a controller test. Ensure that the controller's actions work correctly without requiring a live database.			
• PA 0905: Task: Implement a repository in an MVC project to manage interactions between the business logic and the data access layer, keeping the code modular and easy to test.			
• PA 0906: Task: Implement an exception handling strategy			

across the application that gracefully handles runtime errors, providing helpful error messages to users while logging the details for developers.			
• PA 0907: Task: Handle exceptions in an MVC application by catching errors at runtime (e.g., during form submission) and preventing the application from crashing. Display user-friendly error messages when an unexpected issue occurs.			
• PA 0908: Task: Implement logging throughout the application to track user activities and monitor potential issues. Use logs to identify and fix bugs that are difficult to reproduce in development.			
• PA 0909: Task: Write unit tests for the MVC components, including model classes and controllers. Run the tests and locate potential bugs that could affect the functionality of the application.			

• PA 0910: Task: Build an MVC application that includes a robust exception-handling mechanism to ensure smooth recovery from errors without affecting the user experience.			
• PA 0911: Task: Set up logging providers (e.g., Serilog, NLog) in the application using a common logging API. Ensure logs are recorded for critical events and use them to improve application maintenance and debugging.			

SECTION F- PM-09-PS10:

PM-09-PS10 : Manage security aspects of a web application

Scope of Practical Skill

Given a simulated environment where the conditions are typical of those in a working environment including programming standards and guidelines, programming software, required hardware and its components, industry standard software development tools and an integrated development environment (IDE), the learner must be able to:

- PA 1001 Use identity
- PA 1002 Add basic authentication to the web application utilizing several parameters to ascertain who a user is and provide them with appropriate content while utilizing the applications
- PA 1003 Configure and add login functionality
- PA 1004 Add basic authorisation to the web application for performing specific actions, while protecting users and the application from malicious third parties
- PA 1005 Utilize several different authorisation approaches
- PA 1006 Defend web application against security exploits
- PA 1007 Avoid the cross-site request forgery attack

Applied Knowledge

- AK 1001 Authentication tools
- AK 1002 Authorization in tools
- AK 1003 Defending from Attacks
- AK 1004 Use Identity
- AK 1005 Add Authorization

Internal Assessment Criteria

- IA C1001 Prevent users from accessing sensitive material not intended from them or from performing actions which they should not be able to
- IA C1002 Defend web application against security exploits

Instructions

You are required to perform the task below and your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

NAME: <u>Lindokuhle</u> SURNAME: <u>Moloi</u> COMPANY: <u>Progression</u> ID: <u>0003055567089</u>		DATE: <u>11 May 2026</u> TIME: <u>09:00</u>	
EVALUATION CRITERIONS	COMPETEN T	NOT YET COMPETEN T	COMMENTS OR ACTION REQUIRED
Ability of the learner to:			
PA 1001: Task: Implement identity services in the web application to manage user authentication and roles. Use identity to create a system where users can register and log in securely.			
PA 1002: Task: Add basic authentication to the web application by requiring users to log in with a			

username and password. Ensure the application verifies user credentials and provides access to personalized content.			
• PA 1003: Task: Configure login functionality by adding a login page to the application, ensuring that only authenticated users can access certain areas of the site. Use cookies or tokens to maintain the session.			
• PA 1004: Task: Implement basic authorization to control user access to specific actions or pages in the application. For example, allow only admin users to access the admin panel, while regular users can access their profile pages.			
• PA 1005: Task: Implement multiple authorization approaches in the application, such as role-based and policy-based authorization, to grant or restrict access to various parts of the application based on user roles and permissions.			

• PA 1006: Task: Defend the web application against security exploits like SQL injection, XSS (Cross-Site Scripting), and other vulnerabilities by implementing security best practices such as input validation and encoding.			
• PA 1007: Task: Add anti-forgery tokens to forms and API requests to protect the web application from Cross-Site Request Forgery (CSRF) attacks. Ensure that all form submissions are validated for authenticity.			

SECTION G- PM-09-PS11

PM-09-PS11 : Increase the speed of data access and communication to improve performance

Scope of Practical Skill

Given a simulated environment where the conditions are typical of those in a working environment including programming standards and guidelines, programming software, required hardware and its components, industry standard software development tools and an integrated development environment (IDE), the learner must be able to:

- PA 1101 Implement a caching strategy
- PA 1102 Manage state achieving a state of consistency between different requests
- PA 1103 Implement caching in an application
- PA 1104 Use state management technologies to improve the client experience, by providing a consistent experience for the user
- PA 1105 Implement two-way communication allowing the server to notify the client when important events occur
- PA 1106 Use software library that allows server code to send asynchronous notifications to clientside web applications

Applied Knowledge

- AK 1101 Two-Way communication
- AK 1102 Performance
- AK 1103 Caching strategies

Internal Assessment Criteria

- IA C1101 Use state management technologies to improve the client experience, by providing a consistent experience for the user
- IA C1102 Implement two-way communication allowing the server to notify the client when important events occur

- IA C1103 Use framework that allows the abstraction of several different communication protocols into an easy-to-use API, which allows you to easily create a single set of tools on the server and client to facilitate two-way communications

Instructions

You are required to perform the task below and your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

NAME: <u>Lindokuhle</u> SURNAME: <u>Moloi</u> COMPANY: <u>Progression</u> ID: <u>0003055567089</u>		DATE: <u>11 May 2026</u> TIME: <u>09:00</u>	
EVALUATION CRITERIONS	COMPETEN T	NOT YET COMPETEN T	COMMENTS OR ACTION REQUIRED
Ability of the learner to:			
PA 1101: Task: Implement a caching strategy in the web application by identifying frequently accessed data that can be cached to improve performance. Use in-memory caching (e.g., MemoryCache) to store and retrieve this data efficiently.			
PA 1102: Task: Manage application state by implementing			

techniques that ensure consistency between different requests. Use session state or cookies to maintain user-specific information across multiple requests.			
• PA 1103: Task: Implement caching in the application to reduce load times. Configure caching for data retrieved from a database, ensuring that cached data is refreshed appropriately based on changes in the database.			
• PA 1104: Task: Use state management technologies, such as ViewState or session variables, to improve the client experience by preserving user selections and preferences throughout their session on the web application.			
• PA 1105: Task: Implement two-way communication between the server and client by using technologies like SignalR or WebSockets. Allow the server to notify			

the client in real-time when important events occur, such as new messages or updates.			
• PA 1106: Task: Use a software library (e.g., SignalR) that enables server-side code to send asynchronous notifications to client-side web applications. Create a notification system that alerts users to real-time events without requiring page refreshes.			

SECTION H- PM-09-PS12

PM-09-PS12 : Implement web APIs to enable and promote application interaction with external systems

Scope of Practical Skill

Given a simulated environment where the conditions are typical of those in a working environment including programming standards and guidelines, programming software, required hardware and its components, industry standard software development tools and an integrated development environment (IDE), the learner must be able to:

- PA 1201 Introducing web APIs
- PA 1202 Developing a web API
- PA 1203 Calling a web API using server-side code
- PA 1204 Adding actions and call them
- PA 1205 Create services
- PA 1206 Call a web API from server-side code and jQuery
- PA 1207 Use the Web API to implement Representational State Transfer (REST) services in your application

Applied Knowledge

- AK 1201 Web APIs

Internal Assessment Criteria

- IA C1201 Web APIs are implemented to enable and promote interaction with external systems

Instructions

You are required to perform the task below and your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

NAME: <u>Lindokuhle</u> SURNAME: <u>Moloi</u> COMPANY: <u>Progression</u> ID: <u>0003055567089</u>		DATE: <u>11 May 2026</u> TIME: <u>09:00</u>	
EVALUATION CRITERIONS	COMPETEN T	NOT YET COMPETEN T	COMMENTS OR ACTION REQUIRED
Ability of the learner to:			
PA 1201: Task: Introduce web APIs by explaining their purpose and benefits in modern web development. Provide examples of common web APIs and discuss how they facilitate communication between different software applications.			
PA 1202: Task: Develop a simple web API using ASP.NET Core.			

Create a controller with basic CRUD (Create, Read, Update, Delete) actions for managing a resource, such as products or users.			
• PA 1203: Task: Write server-side code to call a web API and retrieve data. Use HttpClient to send requests to the API and handle the responses appropriately.			
• PA 1204: Task: Add additional actions to the existing web API controller (e.g., search, filter) and ensure they can be called via the appropriate HTTP methods (GET, POST, PUT, DELETE).			
• PA 1205: Task: Create services that interact with the web API. Implement a service layer that abstracts the API calls, providing a clean interface for the rest of the application to use.			
• PA 1206: Task: Call a web API from both server-side code and jQuery on the client side. Use jQuery AJAX methods			

to fetch data from the API and update the user interface dynamically.			
• PA 1207: Task: Use the Web API to implement Representational State Transfer (REST) services in your application. Ensure that the API adheres to RESTful principles, including statelessness and proper use of HTTP status codes.			

SECTION I- PM-09-PS13:

PM-09-PS13 : Host and deploy an web application ensuring it is accessible by clients on a wide variety of machines

Scope of Practical Skill

Given a simulated environment where the conditions are typical of those in a working environment including programming standards and guidelines, programming software, required hardware and its components, industry standard software development tools and an integrated development environment (IDE), the learner must be able to:

- PA 1301 Deploy a web application to the cloud
- PA 1302 Upload an Image to cloud Storage
- PA 1303 Host and deploy an MVC application on a web server
- PA 1304 Host and Deploy an MVC application on a public cloud platform
- PA 1305 Utilise services offered by a public cloud platform to improve the capabilities of your web applications

Applied Knowledge

- AK 1301 On-premise hosting and deployment
- AK 1302 Deployment to a web server
- AK 1303 Fundamentals of a public cloud platform and deployment

Internal Assessment Criteria

- IA C1301 Compile code and compress it to setup an application for a production environment
- IA C1302 Set it up and running on a dedicated server
- IA C1303 Host the web application using a dedicated server to contain the compiled application and
serve it to users as a web-based service
- IA C1304 Deploy the web application by compiling project code and then transferred to the hosting server in compliance the requirements of the hosting environment

Instructions

You are required to perform the task below and your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

NAME: <u>Lindokuhle</u> SURNAME: <u>Moloi</u> COMPANY: <u>Progression</u> ID: <u>0003055567089</u>		DATE: <u>11 May 2026</u> TIME: <u>09:00</u>	
EVALUATION CRITERIONS	COMPETEN T	NOT YET COMPETEN T	COMMENTS OR ACTION REQUIRED
Ability of the learner to:			
PA 1301: Task: Deploy a web application to a cloud platform (e.g., Azure, AWS, or Google Cloud) by following the necessary steps to set up the environment, configure settings, and ensure the application runs correctly in the cloud.			
PA 1302: Task: Upload an image to cloud storage (e.g., AWS			

S3 or Azure Blob Storage) by writing server-side code that handles file uploads and stores the images in the designated cloud storage location, ensuring proper access controls.			
• PA 1303: Task: Host and deploy an MVC application on a web server (e.g., IIS, Apache, or Nginx) by configuring the server environment, deploying the application files, and ensuring that the web server is correctly set up to serve the application.			
• PA 1304: Task: Host and deploy an MVC application on a public cloud platform (e.g., Azure App Service, AWS Elastic Beanstalk) by creating the necessary resources, configuring deployment settings, and verifying that the application is accessible over the internet.			
• PA 1305: Task: Utilize services offered by a public cloud platform (such as databases, storage, or messaging			

services) to enhance the capabilities of your web application. Integrate these services into the application to improve performance, scalability, or functionality.			
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LOGBOOK				
Date	Assignment task	Start	Finish	Total Hours

Date	Learner signature	Date	Mentor/supervisor signature	